MindCanvas

AI Knowledge Graph

An AI system for transforming scattered web data into an interactive, queryable, and personalized knowledge graph.

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Abstract & Objectives: Structuring Digital Wisdom

Abstract: The MindCanvas Solution

MindCanvas is an innovative, AI-driven system designed to combat the problem of fragmented digital knowledge. It transforms the vast, unstructured data encountered during web browsing into a coherent, interactive knowledge graph, making information retrieval semantic and intuitive.

LLMs & RAG

Leverages Large Language Models (LLMs), vector embeddings, and Retrieval-Augmented Generation (RAG) to automatically summarize, categorize, and establish meaningful relationships between information points from diverse web sources.

Knowledge Coherence

The core mechanism is transforming raw text into structured nodes and edges, thereby creating a highly personalized and semantically enriched personal knowledge base.

Core Objectives

Our mission is to empower users with tools that transform passive information consumption into active, structured knowledge creation.

Interactive Knowledge Graph

To build an AI platform that instantly transforms browsing data into an interactive, highly queryable knowledge graph, visually representing connected concepts.

Semantic Conversation

Enable users to visualize, search, and naturally converse with their personal knowledge base using RAG, helping them connect concepts, fill knowledge gaps, and deepen understanding.

The Problem: Fragmented Knowledge and Manual Overhead

Knowledge Fragmentation

Modern learners' knowledge acquisition process is highly fragmented, making it difficult to recall information or connect concepts across multiple distinct sources (articles, videos, documentation).

Poor Retention Rates

Research indicates that approximately 87% of digital learners struggle with knowledge retention, forgetting most of what they learned within a few days without an effective system for active recall and contextualization.

Manual Linking Limitations

Traditional Personal Knowledge Management (PKM) tools like Roam and Obsidian are reliant on manual linking. This dependence severely limits the semantic depth, scalability, and efficiency of the knowledge base.

Lack of Personalization

Existing public web-scale knowledge graphs are generic and lack personalization. They are not tailored to reflect the unique, evolving learning journey or specific information needs of an individual user.

Literature Survey: Foundational Technologies

Our system builds upon recent advancements in AI, vector databases, and RAG frameworks to address the limitations of manual knowledge management.

1	Dense Passage Retrieval... (2020)	Karpukhin, et al.	Dense Passage Retrieval (DPR)	Enabled contextual question answering and efficient document summarization.	Not specifically designed for scalable, personalized knowledge curation from browsing history.
2	pgvector (2024)	Herlihy, A.	Vector similarity search for PostgreSQL.	Makes semantic retrieval computationally feasible at large scale within existing relational databases.	Only provides the storage and search layer; requires a full system implementation around it.
3	LangChain (2024)	Chase, H.	Framework for composable RAG pipelines.	Significantly accelerates the development of complex, multi-step AI workflows.	It is a necessary framework, but not a complete, user-facing application on its own.
4	Roam Research / Obsidian (N/A)	Various	Manual note-linking and bi-directional links.	Excellent for intentional, manual knowledge building and journaling.	Lacks semantic depth and struggles with scalability when dealing with thousands of data points.

MindCanvas: Architecture and Key Modules

Core Advantages

Privacy-First Design

All data capture, processing, and analysis occur locally or on a user's secure server instance. No data is ever shared with third-party vendors.

Deep Semantic Linking

The system moves beyond simple keyword matching, using vector embeddings to retrieve and link content based on conceptual meaning.

Key Modules

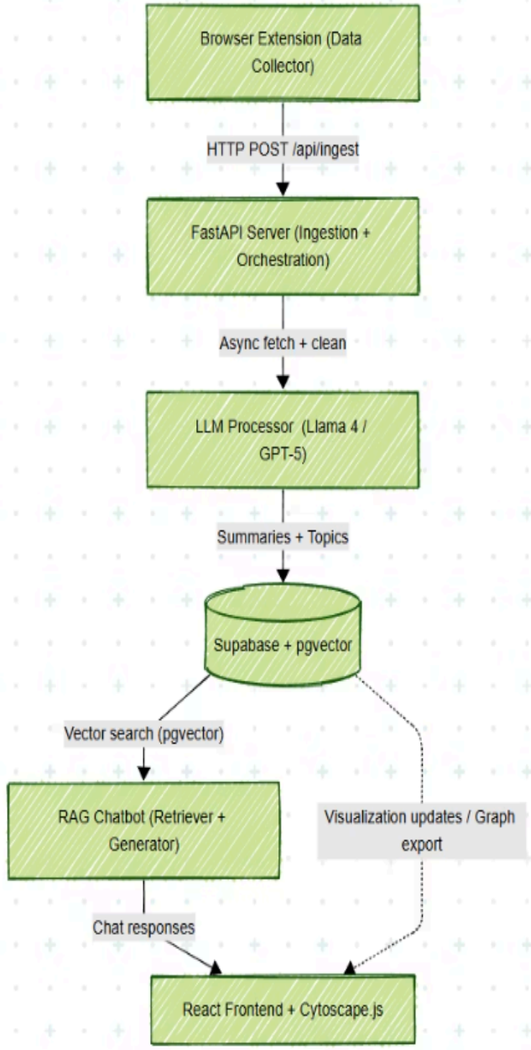
AI Content Analysis

Graph Visualization

Semantic Search

RAG Chatbot

Chrome Extension



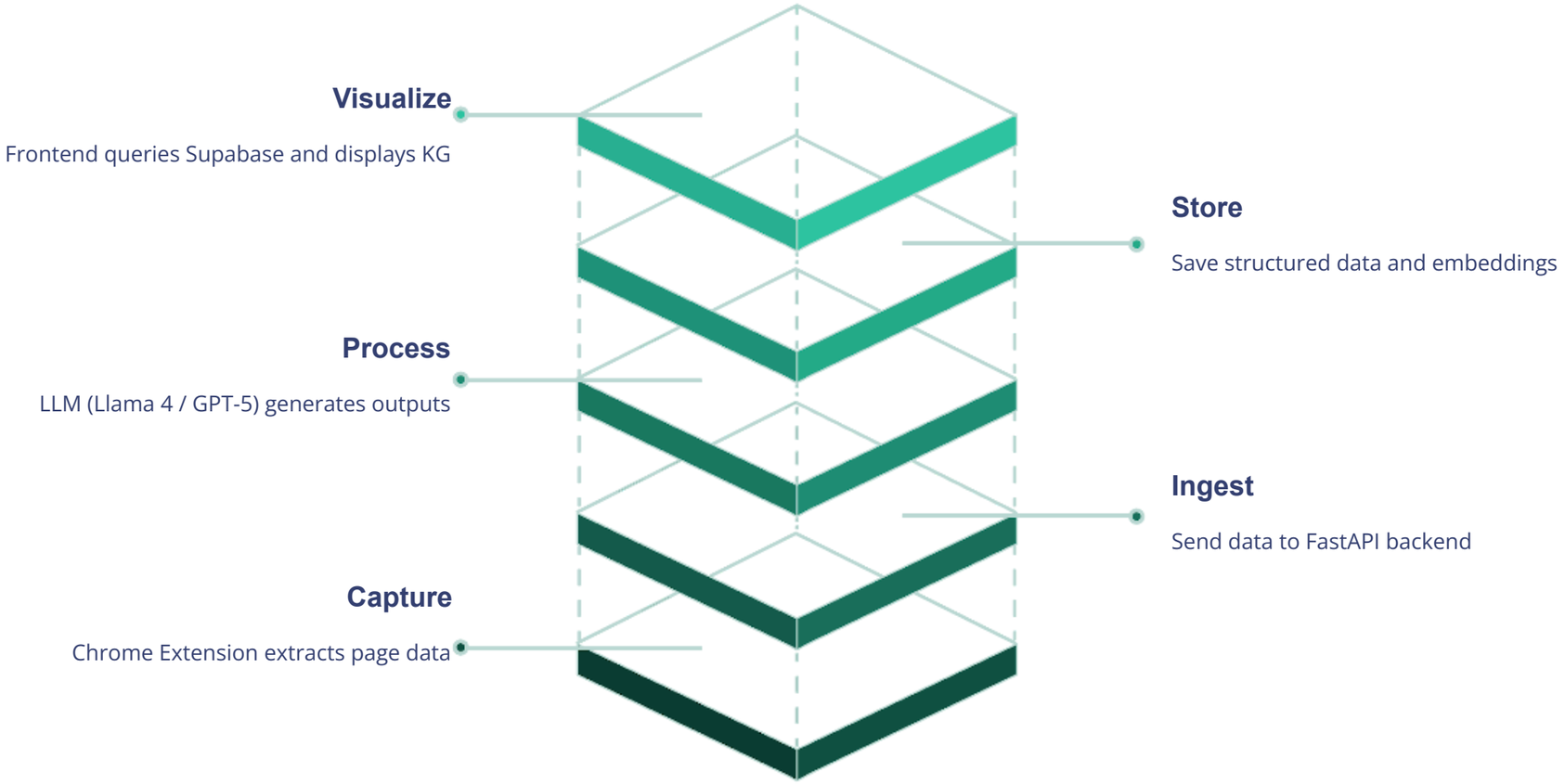
Technical Specifications: Software & Hardware Requirements

A detailed breakdown of the technological stack required to support MindCanvas, ensuring high performance, scalability, and ease of deployment.

AI & Backend	**FastAPI**, **LangChain**, **GPT-5**, **Groq**	Orchestration of complex RAG pipelines, high-performance API serving, and LLM processing.
Database	**Supabase** (**pgvector**)	Efficient semantic vector storage, management of embeddings, and robust data retrieval.
Frontend	**React 18**, **Cytoscape.js**, **Framer Motion**	Creation of the interactive graph visualization interface, a responsive user experience, and smooth animations.
Containerization	**Docker**	Ensuring environment consistency, simplifying deployment across different environments, and enhancing system portability.
Cloud Hosting	**AWS EC2/ECS**	Providing scalable cloud infrastructure necessary for enterprise-grade reliability and handling potential growth in user data.
Browser Ext.	**Manifest V3**	Enabling privacy-first local data capture while adhering to modern browser security standards.

System Architecture & Dataflow

The MindCanvas architecture is designed for low-latency retrieval and high-precision knowledge extraction, integrating seamlessly from the browser to the knowledge graph visualization.



This modular setup ensures that the critical steps of data capture, AI analysis, storage, and visualization are distinct and optimized for their respective tasks, creating a robust and scalable knowledge ecosystem.

Core Algorithms: Precision Knowledge Extraction

The system relies on specialized algorithms for highly accurate information extraction, semantic embedding, graph construction, and grounded Q&A.

LLM Processing for Extraction

We use powerful models (Llama 4 70B primary, GPT-5 fallback) constrained by JSON schemas. This forces the LLM to output accurate, structured data, including summaries, 3-7 key topics (nodes), and a quality assessment score.

Automated Graph Construction

Edges (links) are automatically created between any two content nodes if they meet one of two criteria: (a) they share at least two extracted key topics, OR (b) their respective vector embeddings have a cosine similarity score greater than 0.7.

High-Dimensional Embeddings

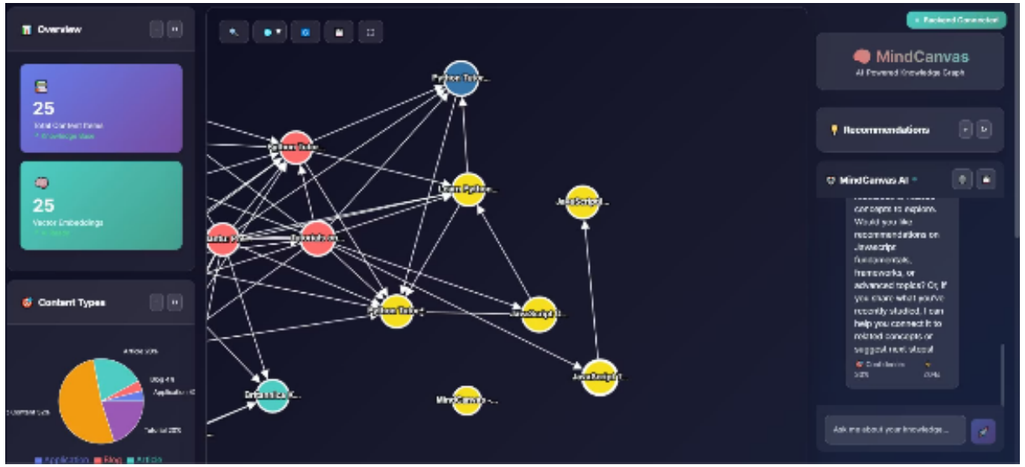
All titles and extracted summaries are converted into 1,536-dimensional vectors using the `text-embedding-3-small` model. This dense representation allows for precise semantic retrieval.

RAG Chatbot Protocol

User queries are first embedded. The system then performs a vector similarity search to retrieve the top-5 most relevant context nodes from the database. These contexts are bundled into a grounded prompt for the generator LLM to produce accurate, context-aware answers.

Result Analysis & Performance Metrics

MindCanvas demonstrates strong performance across core metrics, validating the approach to automated knowledge structuring and retrieval.



Performance Metrics

81%

F1-Score

Combined metric for Topic Extraction (Precision: 0.84, Recall: 0.79).

81%

NDCG@10

Normalized Discounted Cumulative Gain, showing high relevancy in the top 10 retrieval results.

74%

MRR@5

Mean Reciprocal Rank, demonstrating high accuracy in ranking the most relevant document as the first result.

Ablation Study Insight: Llama 4 provided a significant speedup, being 2.6x faster than GPT-5 (1.1s vs 2.9s per page processed) with only a minor, acceptable drop in extraction precision.

Conclusion and Future Directions

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Conclusion: Redefining Knowledge Interaction

- MindCanvas successfully redefines digital knowledge interaction by converting fragmented browsing data into a structured, queryable, and highly intelligent knowledge graph.
- The privacy-first, AI-driven architecture effectively bridges the gap between passive content consumption and active, structured understanding, confirmed by high-quality extraction (0.84 precision) and retrieval (0.74 MRR) metrics.

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Future Work: Expanding the Ecosystem

- Development of personalized learning path recommendations based on identified knowledge gaps within the user's graph.
- Implementation of cloud synchronization across multiple devices with robust, zero-knowledge, end-to-end encryption.
- Integration with existing Personal Knowledge Management (PKM) tools such as Notion, Obsidian, and Roam Research via API access